Response Letter with Annotations

Wu, L., Wang, D., & Evans, J. A. (2019). Large teams develop and small teams disrupt science and technology. Nature (London), 566(7744), 378–382. https://doi.org/10.1038/s41586-019-0941-9

Annotations

1. Purposes
    a. Motivation (one-sentence summary): The study examines how team size influences whether scientific and technological work tends to be disruptive or focused on developing existing ideas.
    b. Specific Rationale (several bullet points):
        - The article start from the idea that science is increasingly done by large teams instead of small teams or solo researchers, but we still don't know how this affects the type of knowledge produced.
        - They argued that previous studies focus on citations (impact), but citations only show how much attention a study receives, not whether it creates new ideas or builds on existing ones. So, they introduce a new idea that can difference them. If a study creates new directions, it is disruptive; if it builds on and improves existing knowledge, it is developmental.
        - To measure this, they build a disruption score (D) using citation patterns. They looked at how future papers cite a study. If they cite it alone, it is disruptive, if they cite it with older references, it is developing.
        - Then, they collected very large datasets from three areas: Scientific papers (Web of Science), Patents, and Software (GitHub)
        - For each work, they measured the team size, the disruption score (D) and the citations
        - First, they tested the main relationship, they compared team size and disruption. They find that as team size increases, disruption decreases, but citations increase with team size.
        - Then, they compare the top 5% most disruptive and most cited papers to understand how team size relates to extreme outcomes. They understood that small teams are more likely to produce highly disruptive work and large teams are more likely to produce high-impact but incremental work
        - After that, they check whether this pattern holds across different contexts, such as across fields and time periods, and indicate that the same pattern appears almost everywhere.
        - They then tested whether the observed relationship is simply driven by differences in the type of research produced by small and large teams. To do this, they control for theoretical versus empirical work and compare

review articles with original research articles. Even after these controls, the results remain the same, showing that the relationship between team size and disruption is not explained by differences in research type.

- They also tested if it is about people instead of team size. They control for authors, topics, and year to know that, but even the same researcher produces less disruptive work in larger teams
- Next, they tryed to explain why this pattern occurs by examining how teams use prior knowledge. They find that small teams tend to build on older and less popular references, whereas large teams are more likely to rely on recent and widely cited work.
- Finally, they examine recognition over time. They suggest that work produced by large teams tends to receive attention quickly, whereas work from small teams often gains recognition later but can have more transformative impact.
- As a conclusion, they argue that small teams tend to disrupt science, while large teams tend to develop it. Both are necessary for a healthy research system, but current scientific practices may overemphasize large teams.

c. Research Questions and/or Hypotheses

- Identify IV, DV, mediator, moderator
    1. **Dependent Variable: Disruption (D)** - degree to which a paper, patent, or software is disruptive vs. developmental
    2. **Independent Variable: Team size** - number of authors/inventors/contributors
    3. **Mediator: Search behavior** - reference age (older vs. recent work) and reference popularity (less cited vs. highly cited work)
    4. **Moderator: Impact (citations level)** – the relationship between team size and disruption become stronger for high-impact work
- Identify specific relationships among variables
    1. Larger team size is associated with lower disruption (more developmental work).
    2. Larger team size is associated with higher citations.
    3. Small teams are more likely to produce highly disruptive work.
    4. Large teams are more likely to produce high-impact but developmental work.
    5. Small teams tend to use older and less popular references, which is associated with higher disruption.
    6. Large teams tend to use recent and popular references, which is associated with lower disruption.
    7. Work from large teams receives faster recognition, while small-team work shows delayed but potentially transformative impact.

2. Method

a. Research Design

- The study uses citation network analysis based on large-scale observational data.

- It constructs citation networks in which nodes represent papers, patents, or software projects, and edges represent citations or knowledge links between them.
- Using these networks, the authors calculate a disruption score (D) based on how future work cites a focal study and its references. They then compare disruption, impact (citations), and team size across datasets from science, technology, and software.
- Data source:
    1. Web of Science (scientific papers and citations)
    2. US Patent and Trademark Office (patents and citations)
    3. GitHub (software projects and fork networks)

b. Sample
- It is a large-scale, non-probability sample based on existing databases (secondary data).
- The data are not representative of a population in a statistical sense, but are extremely comprehensive within each domain.

c. Measures
- **Disruption (D):** Measured using citation patterns in the network, calculated based on how future papers cite the focal work and its references. The score ranges from −1 (developmental) to +1 (disruptive):
    1. Papers that cite only the focal work: increase disruption
    2. Papers that cite the focal work and its references: indicate development
- **Team size:** Measured as the number of authors (papers), inventors (patents), or contributors (software projects)
- **Impact:** Measured as the total number of citations a paper, patent, or software project receives
- **Reference age:** Measured as the average age of the references cited by a work (how old the knowledge is)
- **Reference popularity:** Measured as the number of citations of the references cited by a work (how popular the prior work is)

d. Analysis
- Constructed citation networks to calculate disruption (D)
- Compared team size, disruption, and citations across datasets
- Analyzed top 5% extreme cases (most disruptive and most cited work)
- Tested robustness across fields and time periods
- Ran regression models controlling for year, topic, and author (including author fixed effects)
- Conducted additional analyses to examine search behavior (reference age and popularity) and timing of recognition

3. Results and Conclusions

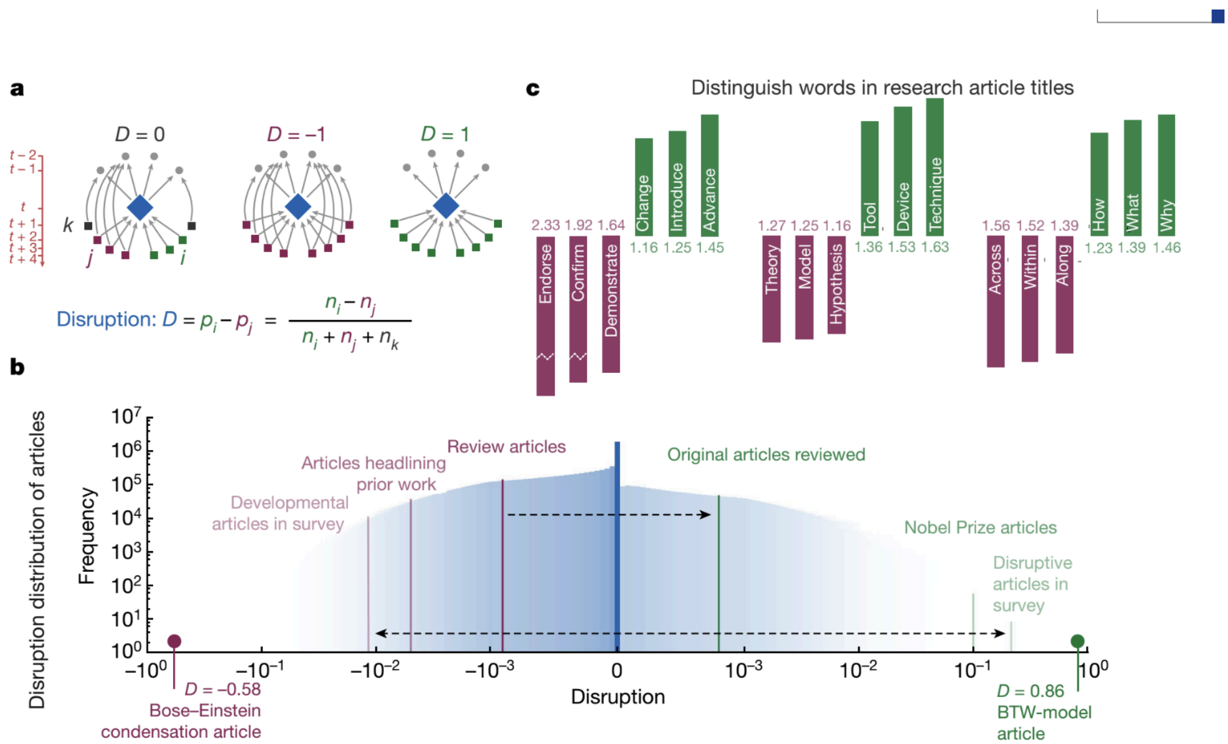
a. Answers to RQs and Hypothese
- **“We predict that work by small teams will be more disruptive than work by large teams”:** The results strongly confirm this prediction: smaller teams disrupt science and technology, whereas larger teams develop existing ideas, with disruption decreasing as team size increases.

- **How does team size affect whether work disrupts or develops science and technology?** The study indicate that smaller teams disrupt science and technology, whereas larger teams develop existing ideas, with disruption decreasing as team size increases.
- **How does team size relate to impact (citations)?** The results show that larger teams receive more citations, meaning higher impact, but this impact is mainly associated with developing work rather than disruptive work.
- **How do small and large teams differ in how they search and assemble knowledge?** The authors find that small teams search more deeply into the past, building on older and less popular work, whereas large teams build on more recent and popular developments.
- **How do small and large teams differ in attention over time?** The results show that work from large teams receives attention immediately, while contributions from small teams succeed further into the future, often with delayed recognition.
- **Are differences between small and large teams explained by other factors or by team size itself?** The conclusion is that differences are not fully explained by topic, research design, or authors, and that most of the effect occurs at the level of team size itself.

b. Include Key Figures/Tables

- Fig 1: Quantifying disruption - Figure 1 presents the construction and validation of the disruption measure (D) through three different panels.
    1. In panel a, the authors use schematic network diagrams to illustrate how the disruption score is calculated based on citation patterns. The focal paper is represented in the center, and the figure shows three possible situations: when future papers cite only the focal paper, when they cite both the focal paper and its references, and when they cite only the references. These different patterns correspond to different values of disruption, ranging from -1 (developmental) to +1 (disruptive). While this panel is useful to conceptually explain the variable, it is not very intuitive at first glance and requires support from the text to be fully understood.
    2. Panel b shows the distribution of disruption scores across a large number of articles. The X-axis represents the disruption score, ranging from -1 to +1, and the Y-axis represents the frequency of articles. The figure shows that most articles are concentrated around zero, suggesting that most work is neither highly disruptive nor purely developmental. The authors also include examples, such as Nobel Prize-winning articles and well-known studies, to illustrate how highly disruptive work tends to appear on the right side of the distribution. This panel helps validate the measure, but it relies on interpretation and prior knowledge of the examples.
    3. Finally, panel c provides an additional validation by showing differences in the language used in article titles. Words associated with disruptive work (“introduce,” “advance”) are contrasted with

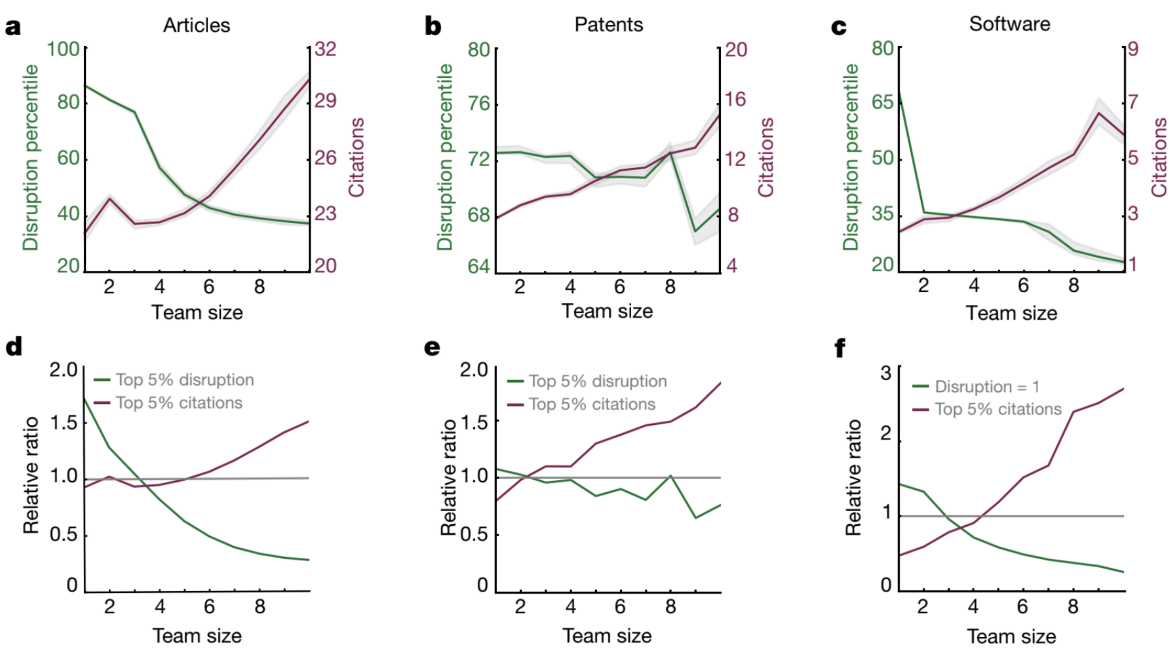
words associated with developmental work (“confirm,” “demonstrate”). This suggests that the disruption score captures meaningful differences in the type of contribution. However, this panel is more interpretative and less directly connected to the main analysis.



- Fig 2: Figure 2 presents the main relationship between team size, disruption, and impact across three domains: articles, patents, and software.
    1. In panels a, b, and c, the X-axis represents team size, while the Y-axes represent disruption percentile (left axis, in green) and citations (right axis, in purple). Across all three panels, the figure shows a consistent pattern: as team size increases, disruption tends to decrease, while citations increase. This visual representation clearly supports the main argument of the study, making it easy to see the contrasting roles of small and large teams. However, the smooth trends shown in the lines may oversimplify the data and make the relationship appear more uniform than it actually is, potentially hiding variation within groups.
    2. Panels d, e, and f further examine this relationship by focusing on extreme cases. In these panels, the X-axis still represents team size, while the Y-axis shows relative ratios. The green line represents the likelihood of producing highly disruptive work (top 5% disruption), and the purple line represents the likelihood of producing highly cited work (top 5% citations). These panels

reinforce the main finding: smaller teams are more likely to produce highly disruptive work, while larger teams are more likely to produce highly cited work. The use of relative ratios helps highlight these differences more clearly. However, the interpretation of these panels may be less intuitive, especially for readers unfamiliar with relative measures, and still requires some guidance from the text.

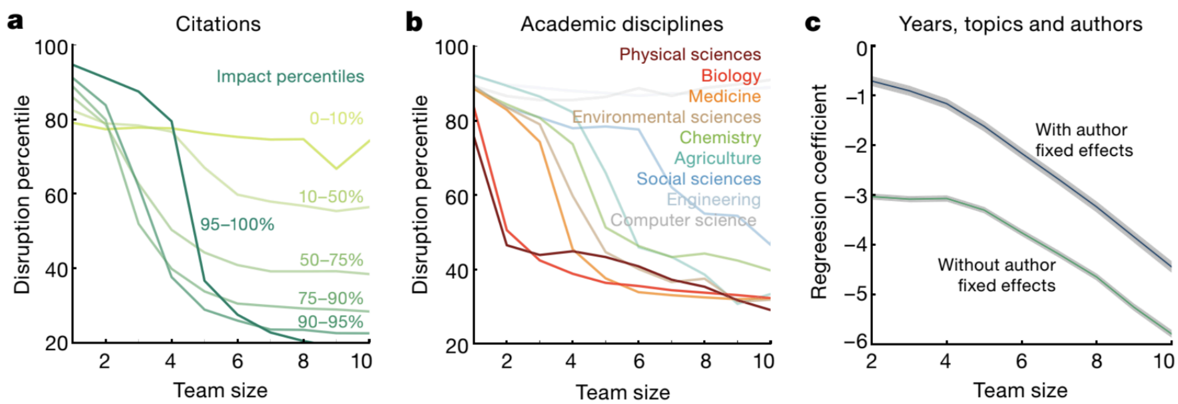
3. Overall, Figure 2 is effective in visually communicating the core findings of the study, particularly through the consistent patterns across different domains. At the same time, its reliance on aggregated trends and relative measures may limit the visibility of individual variation and make some aspects of the visualization less immediately accessible.

- 

- Fig 3: Small teams disrupt across impact levels and fields
  1. Figure 3 a shows that the pattern of small teams being more disruptive happens consistently across different levels of impact. The X-axis represents team size and the Y-axis represents disruption. Each line corresponds to a group of articles with different impact percentiles. All lines decrease as team size increases, meaning that even among highly cited articles, larger teams are still less disruptive. Therefore, the effect does not depend on the level of impact.
  2. Figure 3 b shows that this pattern also holds across different academic disciplines. The X-axis represents team size and the Y-axis represents disruption, and each line represents a different

field, such as biology, medicine, or engineering. Again, all lines decrease as team size increases, indicating that larger teams are less disruptive in all fields. This suggests that the effect is not limited to a specific discipline.

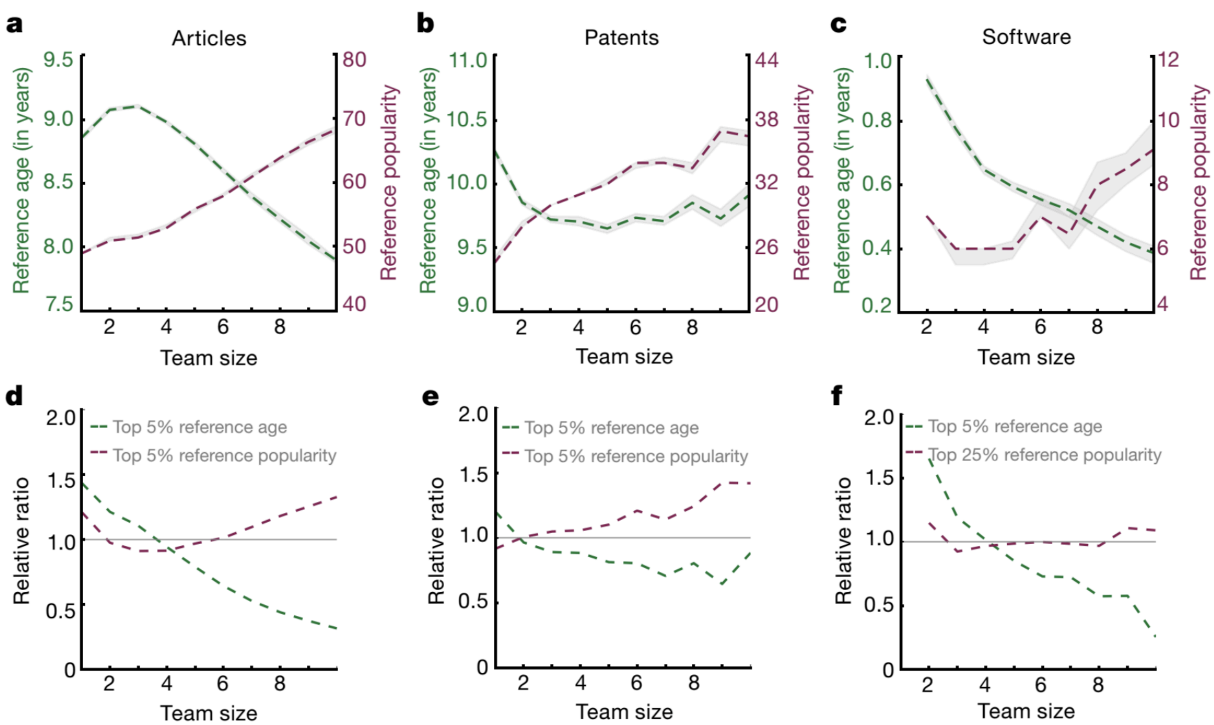
3. Figure 3 c is whether this relationship is explained by differences between authors. The X-axis represents team size and the Y-axis shows the regression coefficient. The figure compares models with and without author fixed effects. In both cases, the lines decrease as team size increases, showing that even the same researcher produces less disruptive work when working in larger teams. This suggests that the effect is driven by team size itself rather than by individual differences.



- Fig 4: Small and large teams behave differently in their search through past work - This figure connects the main result to a possible explanation, showing differences in how teams use references.
  1. Figure 4 a shows how reference age changes with team size in articles. The X-axis represents team size and the Y-axis represents reference age. The line shows that as team size increases, the age of the references decreases. This means that larger teams tend to use more recent work, while smaller teams rely more on older knowledge.
  2. Figure 4 b shows the same pattern for patents. As team size increases, reference age decreases, indicating that larger teams build more on recent work, while smaller teams search deeper into the past.
  3. Figure 4 c shows this relationship for software projects. Again, the X-axis is team size and the Y-axis is reference age. The pattern is consistent, with larger teams using more recent references and smaller teams using older ones. This reinforces the idea that search behavior differs by team size.
  4. Figure 4 d, e an f focuses on extreme cases for articles. The X-axis represents team size and the Y-axis shows the relative ratio. The

green line represents the top 5% of reference age (older references), and the purple line represents the top 5% of reference popularity. For d related to article, as team size increases, the likelihood of using older references decreases, while the use of more popular references increases.

5. Figure 4 e shows the same analysis for patents. The pattern is similar, with larger teams being more likely to rely on popular references and less likely to use older ones.
6. Figure 4 f shows this pattern for software. The green line decreases and the purple line increases, indicating that larger teams rely more on popular and recent work, while smaller teams rely more on older and less popular references.



c. Authors' Conclusions

- Small teams and large teams play different but complementary roles in science and technology. Small teams are more likely to disrupt existing knowledge by introducing new ideas and directions, while large teams are more likely to develop existing ideas by refining and extending them.
- These differences are largely explained by how teams search and use prior knowledge. Small teams tend to build on older and less popular work, whereas large teams rely on recent and widely cited work, leading to different types of contributions.

- The study shows that impact and disruption are not the same. Large teams tend to receive more citations and gain attention quickly, but small teams are more likely to produce transformative contributions that may be recognized later.
- Both small and large teams are essential for a healthy research ecosystem, as they contribute in different ways: small teams open new directions, and large teams consolidate and expand them.
- However, the increasing dominance of large teams suggests that current scientific practices may overemphasize development over disruption, highlighting the need for policies that also support small-team research.

4. Discussion
    a. Implications
        - The article shows that impact (citations) and innovation (disruption) are not the same, challenging evaluation systems that rely mainly on citation counts to assess scientific contributions.
        - It highlights that team size shapes the type of knowledge produced, with small teams driving novel directions and large teams supporting incremental development.
        - The findings suggest that the current scientific system, which increasingly favors large teams, may overemphasize development over disruption, potentially limiting radical innovation.
        - The study also contributes to theories of knowledge production and team science, showing that differences in outcomes are linked to how teams search and use prior knowledge.
        - Overall, it argues that a balanced research ecosystem is necessary, where both small and large teams are supported because they play complementary roles.
    b. Limitations
        - It is observational, meaning that causal relationships cannot be definitively established between team size and disruption.
        - Although the datasets are large, they rely on secondary data (papers, patents, software) and may not capture all aspects of knowledge production (e.g., informal collaboration, failed projects).
        - The measure of disruption is based on citation patterns, which may not fully capture all forms of innovation or scientific contribution.
        - Differences across fields are considered, but the study may still overlook contextual factors such as institutional settings or funding structures.
        - The analysis focuses on published outputs, which may bias results toward successful or visible research.
    c. Future Directions
        - Future research could explore causal mechanisms more directly, for example, by examining how team processes influence decision-making and innovation.
        - Studies could investigate how funding structures and institutional incentives shape the balance between small and large teams.

- Future work could examine whether similar patterns exist in other contexts, such as different countries, disciplines, or emerging research fields.
- Researchers could explore additional measures of innovation beyond citations, incorporating qualitative or content-based indicators.
- It would also be valuable to study how to design policies that better support small-team research, preserving its role in generating disruptive ideas.

--- Page Break Here ---
Response Letter (~600 words)

- Your assessment of the study

I think this study makes a strong contribution to the understanding of the scientific production and evolution. One of its main strengths is the scale of the data, as the authors analyze more than 65 million projects across several decades. This large-scale approach allows them to identify consistent patterns that would not be visible in smaller studies. In addition, the use of multiple domains (science, technology, and software) strengthens the generalizability of the findings.

The study is also methodologically rigorous. The combination of citation network analysis, bibliometric analysis, and regression models with controls demonstrates careful attention to how they build the internal validity. This is particularly important because the authors show that the relationship between team size and disruption is not simply due to differences in research type, topic, or individual researchers. Instead, team size itself appears to play a meaningful role.

Finally, the results have important implications. The distinction between small teams as drivers of disruption and large teams as drivers of development offers a compelling framework for thinking about how science progresses. It also raises critical questions about current scientific practices, especially the increasing dominance of large teams and the potential consequences for innovation.

About the figures:

Overall, the figures are helpful to understand the main argument of the study, but they vary in clarity and effectiveness.

Figure 1 is confusing, as it combines different types of information in the same figure. Panels a, b, and c represent different ideas (conceptual explanation, distribution, and validation), which makes it difficult to follow without reading the text. It could be three different figures.

Figure 2 is the clearest and most effective, as it directly shows the relationship between team size, disruption, and citations. The use of simple lines makes the main finding easy to understand, although the smooth trends may hide variation in the data.

Figure 3 reinforces the main result by showing that the pattern holds across impact levels, disciplines, and controls. However, the multiple overlapping lines and similar colors make it harder to interpret quickly.

Figure 4 helps explain the mechanism behind the results by showing differences in how teams use references. While it is useful, the connection between these variables and disruption is not fully clear from the visualization alone.

I think the figures are effective in presenting the main relationships, but I still had to read the text to fully interpret them. In my opinion, the authors tried to include as much information as possible in the figures, which is not always ideal. More intuitive and simple visualizations, as well as more clear representations of variability, could improve clarity and understanding.

- Weaknesses

First, the design is observational, which means that causal relationships cannot be definitively established. Although the authors control for many factors, it is still possible that unobserved variables influence both team size and the type of contribution.

Second, the disruption measure relies entirely on citation patterns. While the authors validate this measure, citations may still be an imperfect data for innovation. For example, citation behavior can be influenced by social, institutional, or strategic factors, and not all important contributions are reflected in citation networks.

Third, the study does not directly observe team processes. The explanation for why small and large teams differ is based on indirect evidence, such as reference age and popularity, rather than direct observation of decision-making, collaboration dynamics, or creativity within teams.

Finally, although it considers differences across fields and time, it does not deeply explore contextual factors such as funding structures, institutional pressures, or disciplinary norms, which may also shape how teams produce knowledge.

- Other notes

One interesting extension of this study would be to connect its findings to research on innovation, risk, and organizational behavior. For example, the idea that small teams take more risks and explore less popular ideas aligns with theories of exploration versus exploitation. It would be valuable to further investigate how institutional incentives, such as funding and publication pressures, influence whether researchers choose to work in small or large teams.

Another possible direction would be to apply this framework to areas such as misinformation, health communication, or policy research, where understanding the balance between disruptive and developmental knowledge could be particularly relevant. Overall, the study opens important path for future research and provides a useful framework for rethinking how scientific contributions are evaluated and supported.